

NIMCET 2027 SYLLABUS

Prepared by NIMCET Aspirant

Mathematics

Algebra

- Logarithmic
- Quadratic equations and Inequalities
- Progression
- Exponentials
- Binomial Theorem
- Determinant
- Matrix
- Set Theory
- Permutation and Combination
- Probability
- Complex numbers
- Mathematical Logic (Tautology, Contradiction, Truth Table, Connectives)

Trigonometry

- Measurement of Angles
- Trigonometric ratios and functions
- Trigonometric equations
- Properties of a Triangle
- Inverse Trigonometry Function
- Height and distance

Calculus

- Function
- Limit
- Continuity
- Differentiability
- Differentiation
- Application of Derivative
- Indefinite integrals
- Definite Integral
- Area Under Curve
- Differential Equation

Coordinate Geometry

- Rectangular Cartesian System
- Straight line
- Pair of Straight line
- Circle
- Parabola
- Hyperbola
- Ellipse
- Statistics

Quantitative Aptitude

- Data Interpretation
- Data Sufficiency
- Numerical Reasoning
- Data Visualization

Reasoning

- Coding Decoding
- Blood Relation
- Seating Arrangement
- Direction
- Input Output
- Syllogism
- Alphanumeric Series
- Mirror Images
- Statements and Conclusions/arguments
- Puzzles

Computer Awareness

Computer Basics

- Organization of a computer
- Central Processing Unit (CPU)
- Structure of instructions in CPU
- Input/output devices
- Computer memory
- Back-up devices

Data Representation

- Representation of characters, integers, and fractions
- Binary and hexadecimal representations
- Binary arithmetic: addition, subtraction, multiplication, division
- Two's complement arithmetic
- Floating point representation of numbers
- Boolean algebra

Computer Hardware

- Input Devices: Keyboard, mouse, scanner, etc.
- Output Devices: Monitor, printer, speakers, etc.
- Storage Devices: Hard drives, SSDs, USB drives, etc.
- Memory: RAM, ROM, cache, etc.

Computer Software

- Operating Systems: Windows, macOS, Linux, Android, etc.
- System Software: Utility programs, device drivers
- Application Software: Basic concepts & tools

Internet and Email

- Web Browsing - Understanding how the internet works and navigation
- Email - Sending, receiving, managing emails
- Online Security - Basic awareness of online threats and safety measures

General English

- Comprehension
- Vocabulary
- Basic English Grammar
- Word Power
- Synonyms and Antonyms
- Meaning of words and phrases
- Technical writing